INCH-POUND

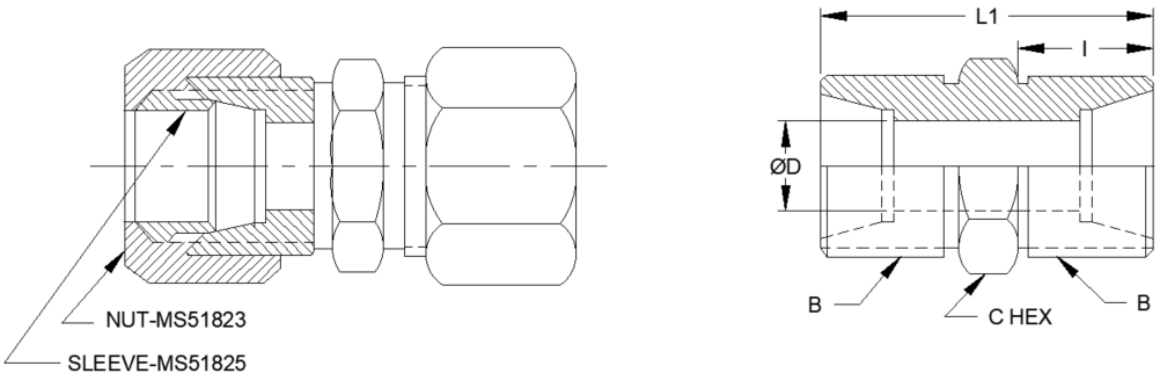
MS51814F
w/AMENDMENT 2
21 November 2019
SUPERSEDING
MS51814F
w/AMENDMENT 1
10 October 2018

DETAIL SPECIFICATION SHEET

NIPPLE, TUBE, REGULAR HEX, FLARELESS TYPE, HYDRAULIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



See notes at end of figure.

FIGURE 1. Nipple, tube.

AMSC N/A

FSC 4730

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Dash number		Tube OD Nom inches (mm)	B Straight thread Class 2A	C Hex Flat Nom (see note 6)	D dia. Drill inches (mm)
Assembly	Body (see note 5)				
1	21	.1250 (3.175)	.3125 - 24UNF	7/16	.093 (2.36)
2	22	.1875 (4.763)	.3750 - 24UNF	7/16	.125 (3.18)
3	23	.2500 (6.350)	.4375 - 20UNF	1/2	.203 (5.16)
4	24	.3125 (7.936)	.5000 - 20UNF	9/16	.234 (5.94)
5	25	.3750 (9.525)	.5625 - 18UNF	5/8	.281 (7.14)
6	26	.5000 (12.700)	.7500 - 16UNF	13/16	.422 (10.72)
7	27	.6250 (15.875)	.8750 - 14UNF	15/16	.500 (12.70)
8	28	.7500 (19.050)	1.0625 - 12UN	1-1/8	.656 (16.66)
9	29	.8750 (22.225)	1.1875 - 12UN	1-1/4	.718 (18.24)
10	30	1.0000 (25.400)	1.3125 - 12UN	1-3/8	.875 (22.23)
11	31	1.2500 (31.750)	1.6250 - 12UN	1-11/16	1.093 (27.76)
12	32	1.5000 (38.100)	1.8750 - 12UN	2	1.344 (34.14)
13	33	2.0000 (50.800)	2.5000 - 12UN	2-5/8	1.813 (46.05)

Dash number		I inches (mm) ±.15 (0.38)	L_1 inches (mm) ±.02 (0.51)
Assembly	Body (see note 5)		
1	21	.375 (9.53)	1.02 (25.91)
2	22	.422 (10.72)	1.11 (28.19)
3	23	.453 (11.51)	1.18 (29.97)
4	24	.453 (11.51)	1.18 (29.97)
5	25	.469 (11.91)	1.24 (31.50)
6	26	.562 (14.27)	1.42 (36.07)
7	27	.625 (15.88)	1.61 (40.89)
8	28	.688 (17.48)	1.81 (45.97)
9	29	.688 (17.48)	1.81 (45.97)
10	30	.688 (17.48)	1.81 (45.97)
11	31	.688 (17.48)	1.89 (48.01)
12	32	.688 (17.48)	1.96 (49.78)
13	33	.688 (17.48)	2.11 (53.59)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Assemblies are only furnished to this specification (body, nuts, and sleeves). Bodies are not to be stocked, stored, or issued.
6. Across flat widths shall fit standard wrench openings, see ASME B18.2.2.
7. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.

FIGURE 1. Nipple, tube - Continued.

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REQUIREMENTS:

Nipple, tube, bulkhead shall be as specified on figure 1 and in table I.

Materials shall be in accordance with MIL-DTL-18866.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

Finishes shall be as specified in table I.

All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

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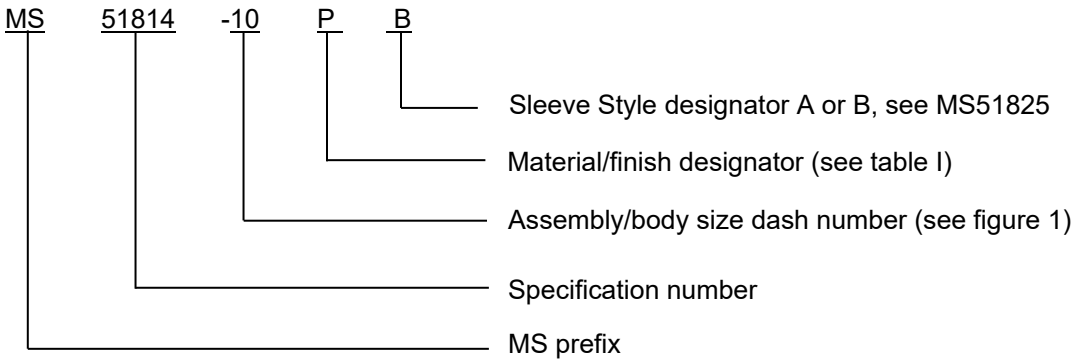
TABLE I. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ 7/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 4/
R	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/ 5/
SS	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7, free machining alloy UNS S30300 method 1, type 2 or 4.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 6/ 8/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 6/ 8/
Z	Steel	Zinc plating in accordance with ASTM B633, type VI, Fe/Zn 8. 4/
ZC	Steel	Zinc may be any zinc plating from PIN codes J and R with a colored chromate coating. 4/ 5/

1/ Embrittlement test need not be run.
2/ Hexavalent chromium free. Finish shall be RoHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook.
5/ These finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended for Army ground systems.
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.
7/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA. The use of cadmium plating for Army vehicles is prohibited unless otherwise explicated stated in the contract as a requirement.
8/ Titanium shall not be used in oxygen or potable water systems.

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Part or Identifying Number (PIN): The PIN consists of the letters “MS”, the specification number, a dash, number for nipple tube assembly/body size, a letter for material finish designator, and a letter for a sleeve style.



PIN example: MS51814-10P indicates a nipple, tube assembly, 1.000 inch (2.540 mm) tube OD, with 1.3125 - 12UN-2A threads, steel, zinc phosphate finish, sleeve designator B.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

The use of cadmium plating for Army vehicles is prohibited unless otherwise explicated stated in the contract as a requirement.

Class I and II ozone depleting substances (ODS) shall not be used in MS51814 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Table I provides a detail reference data listing for MS39201 to MS51814. This table contains coordination of equivalent tube sizes, in both compression and hydraulic type nipples. Assembly replacement is authorized.

TABLE II. Assembly equivalency (for information only).

PIN		PIN		PIN	
Compression MS39201	Hydraulic MS51814	Compression MS39201	Hydraulic MS51814	Compression MS39201	Hydraulic MS51814
-1	-1	-5	-5	-10	-9
-2	-2	-6	-6	-11	-10
-3	-3	-8	-7	-11	-12
-4	-4	-9	-8	-12	-13

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

MIL-DTL-16232	SAE-AMS-QQ-P-416	SAE-AMS-STD-595/36251
MS39201	SAE-AMS-STD-595/36076	SAE-AMS-STD-595/36270
MS51823	SAE-AMS-STD-595/36081	SAE-AMS-STD-595/36280
MS51825	SAE-AMS-STD-595/36099	SAE-AMS-STD-595/36293
ASME B18.2.2	SAE-AMS-STD-595/36118	SAE-AMS2417
ASTM B117	SAE-AMS-STD-595/36134	SAE-AMS2486
ASTM B633	SAE-AMS-STD-595/36152	SAE-AMS2488
ASTM F1136/F1136M	SAE-AMS-STD-595/36170	SAE-AMS2700
ASTM MNL37	SAE-AMS-STD-595/36173	SAE-J514
SAE-AMS-C-81562	SAE-AMS-STD-595/36176	T9070-AL-DPC-020/077-2
	SAE-AMS-STD-595/36231	

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2019-080)

Review activities:
Army - AR, MI
Navy - AS, CG, SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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